Instructor Guide

In this activity, students will be taking on the role of reviewers for a foundation which funds projects that fight cancer. They will be responsible for reading six grant proposals and making recommendations on how to spend their foundation's budget.

1) Break students into groups of 3 – 4 and provide them with:
    - 1 set of instructions
    - 1 set of proposals

2) Give groups 30 – 60 minutes (this is flexible and up to you) to discuss the proposals and make a set of funding recommendations. Note that their overall budget ($7,500,000) is too low to fund every project.
    - Students should be prepared to deliver a brief "report" for each proposal explaining why it should be funded (or not). This should be roughly 100 words but doesn't necessarily have to be written down.
    - Advise students that they should be drawing on their subject knowledge from M&L as well as their habits of mind. Some proposals might require a calculator, scratch paper, or some working out to decide on.

3) Have each group send a representative to the board to put up their list of funding recommendations.
    - You can run this wrap-up in a couple different ways. If groups differ in their suggestions, representatives can move between groups to discuss differences of opinion.
    - Or you can have everyone come together to discuss which projects the class will choose to fund.
    - Or you can have each group present to you when they are finished, and let each group be its own separate "organization".

The major roadblock when running this activity is getting students to **actually think carefully** about each proposal. Because it is an open-ended activity, students tend to make quick decisions based on superficial characteristics. Try to make it clear that yes, they should be calculating confidence intervals or probabilities. They should be looking at the codon code and trying to figure out what these sequences mean, or how each of these therapeutics will affect information flow (as in the central dogma).

Note – there aren't "right" or "wrong" answers to this activity. Groups can choose to fund whichever proposals they want. However, applying Molecules & Life subject knowledge along with applicable habits of mind will improve decision-making.

To back this up, make yourself available as a "representative" of any of the grants that students would like to talk more about. On the following pages of this handout are "cheat sheets" with suggestions and prompts for each of the proposals. You can push students to consider all aspects of each proposal with the given hints and suggestions. You can even role play the various characters if you want!

Pancreax PCX-1 screening

Prompt questions:

- If a Pancreax test is positive, what are the odds that the patient has pancreatic cancer?
- How much will it cost (out of pocket) for all these positive tests to get biopsies?

More directly suggest:

- Consider probabilities
- Consider costs
- Consider return on investment

Worth Noting:

- The odds of a New Yorker over 40 developing pancreatic cancer are $\frac{400}{3{,}700{,}000} = \mathbf{1.081} * \mathbf{10^{-4}}$. In a population of 850,000 individuals, we would expect $(\mathbf{8.5} * \mathbf{10^{5}}) * (\mathbf{1.081} * \mathbf{10^{-4}}) \approx \mathbf{92}$ cases
- The odds of a false positive are $\frac{5}{406} = \mathbf{1.23} * \mathbf{10^{-2}}$
- In a population of 850,000 individuals, we would expect $(\mathbf{8.5} * \mathbf{10^{5}}) * (\mathbf{1.23} * \mathbf{10^{-2}}) \approx \mathbf{10{,}455}$ false positives.
- So, if an apparently healthy patient receives a positive test as part of their annual physical, the odds that they have pancreatic cancer are $\frac{92}{(10{,}455+92)} = \mathbf{0.0087}$
- The out-of-pocket cost for a biopsy (and the time/stress) is very high!

Forward Health Advocates

Prompt questions:

- How was this chemical uncovered?
- What is the link between this chemical and lung cancer?
- Do the scientific statements of the writer match with the statements of the analysis company?

More directly suggest:

- Was the survey an experimental or correlational study?
- Can you cause cancer by affecting a protein directly?
- Do the results of the analysis suggest that this VOC causes mutations?

Worth Noting:

- The survey is a correlational study which turned up one significant correlation out of a whole batch of non-significant findings. This is a bit suspicious.
- This VOC appears to bind directly to a protein that signals for cellular replication. This is not cancer and does not affect the underlying genes.
- It could still be bad, though! Petri-dish findings aren't always conclusive, but if exposure to this chemical causes unplanned cell division, that's not a good thing.

Phylogen

Prompt questions:

- Which findings are significant, and which are not?
- Does this vaccine appear to work?
- What is the connection between this vaccine and preventing cancer?

More directly suggest:

- Did being vaccinated reduce the risk of developing cancer in mice?

Worth Noting:

- The vaccine appears to be effective in a mouse model – and this alone might be worth putting some money into next steps.
- Mice who were vaccinated were not less likely to develop cancer than those who were not, but it's not clear that this is disqualifying.
- Other research has shown a link between Epstein-Barr and cancer, so a vaccine doesn't necessarily need to prove this link itself.

Genetechnic

Prompt questions:

- Could this approach work? Can you break oncogenes using CRISPR?
- How does the OncoStop sequence work (you might need a codon code for this)?
- What are the risks of breaking genes in this way?

More directly suggest:

- How many 8-nucleotide sequences are there, and how often do we expect one of these to be repeated across the genome?
- What are the odds of editing other genes by accident?

Worth Noting:

- The idea is broadly a good one – deactivating oncogenes using CRISPR is theoretically possible (if we ignore many, many hurdles).
- But this approach is not going to work. There will be many, many instances of any given 8-nucleotide sequence. So, assuming the delivery of the CRISPR system works, there are going to be lots and lots of off-target edits, some of which will cause damage to other genes.
- OncoCut is fun, but not necessarily a good approach either. It is simply a stop codon iterated across all three potential reading frames (if students get that, they're doing ok). This might break a gene by truncating translation, but you can probably also break an oncogene just by cutting it and letting NHEJ do the rest.
- Specifically injecting a stop codon into the middle of a gene also increases the odds of really breaking any of the off-target cuts that do occur.

Cheminonon

Prompt questions:

- What are the differences between the two offered approaches?
- Which approach is safer? Which seems more likely to reduce tumor growth?

More directly suggest:

- How do these two approaches fight cancer? Describe the mechanisms they are using.
- Which are you more concerned with, killing cancer cells or reducing side effects?

Worth Noting:

- Both approaches could work. Nucleon's approach is newer and not as well tested. Chemionics is basically offering a classic chemotherapy drug.
- Either approach could be defensible. One offers a well-known method with well-known drawbacks. The other is untested, potentially better, but less of a sure thing. Which one students go for (if either) is going to be determined by temperament and appetite for risk.
- It's good for students to think about how to make subjective, emotional decisions with incomplete data.

Healthline NYC

Prompt questions:

- What do the findings of these two studies suggest about the efficacy of the proposed ad campaigns?

More directly suggest:

- You should calculate 95% CI's for these results and see whether there is a difference between pre- and post-viewing attitudes!

Worth Noting:

- Only the "Screen It!" ad campaign has a significant effect. Students might be tempted to throw the full million at this since it's relatively cheap compared to the other requests. But there is no evidence that the "Smoke is no Joke" campaign will be at all successful.
- Students might want to consider why some kinds of behavior are easier to change through an ad campaign than others. Is there any reason why suggesting people wear sunscreen might be an easier sell than suggesting people stop smoking?
- If so, what effect might this have on public health decision-making?